AAMAS 2027 — Blue Sky Ideas Track

Additional Experimental Results

Learning Signed Critic Errors: A Probabilistic Approach to Addressing Function Approximation Errors in Actor–Critic Reinforcement Learning

Anonymous Submission

Additional Experimental Analysis

We present additional experimental results for DDPG, TD3, and SAC on the Gymnasium MuJoCo Hopper-v5 environment across five random seeds. The analysis examines critic value estimates, signed estimation errors, and Gaussian Process Regression (GPR)-based corrections during training.

Critic Value Estimation. Figure 1 compares critic estimates with Monte Carlo (MC) return references. DDPG exhibits substantial early overestimation, while TD3 generally underestimates the reference. SAC shows smaller discrepancies toward later training.

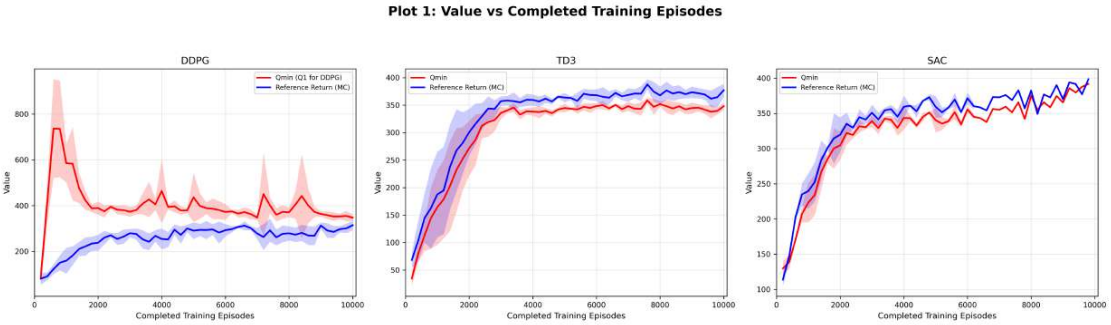


Figure 1: Critic estimates and MC return references for DDPG, TD3, and SAC on Hopper-v5.

Signed Critic Errors. Figure 2 illustrates the direction and evolution of critic estimation errors. DDPG shows predominantly positive discrepancies, while TD3 and SAC exhibit mostly negative errors, highlighting the importance of modeling both overestimation and underestimation.

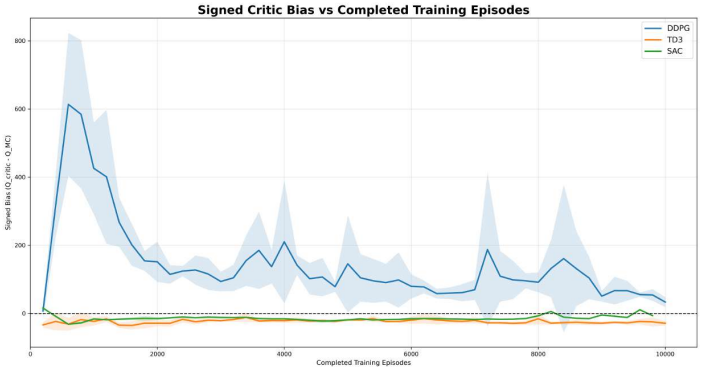


Figure 2: Signed critic–reference discrepancies during training on Hopper-v5.

GPR-Based Error Correction. Figure 3 compares raw and GPR-corrected discrepancies. Corrected values remain closer to zero on the evaluated fitting samples, suggesting that GPR can capture signed residual patterns.

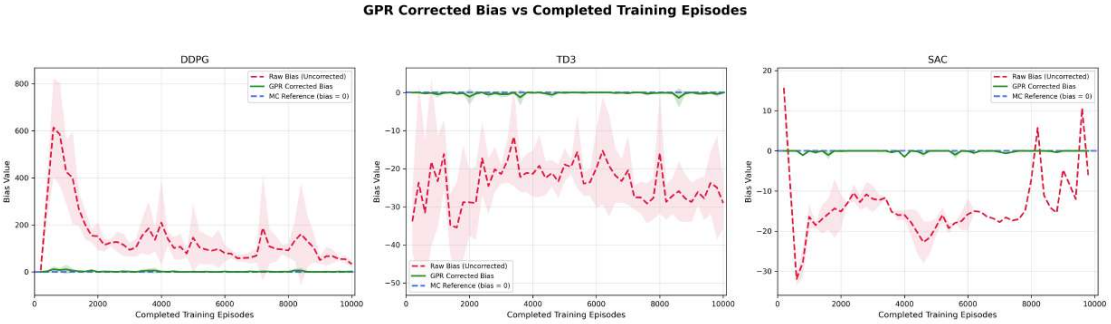


Figure 3: Raw and GPR-corrected critic discrepancies for DDPG, TD3, and SAC on Hopper-v5.

Discussion. These preliminary results motivate probabilistic critic-error correction for actor–critic learning in continuous-control environments. Since GPR corrections are evaluated on fitting samples, further independent

evaluation is needed to establish generalization and improvements in policy performance.